INCH-POUND

MIL-M-71074A (AR)
25 January 1995
(See 6.8)

MILITARY SPECIFICATION

MODEM, VEHICLE MOTION SENSOR: 12561737

This specification is approved for use by the U.S. Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Vehicle Motion Sensor (VMS) Modem used in the Automatic Fire Control System (AFCS) for the M109A6 self-propelled howitzer (SPH) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-795	-	Cushioning Material, Packaging (Flexible Closed Cell Plastic Film for Long Shipping Cycle Applications)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1290

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MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-117	-	Bag, Sleeve, and Tubing - Interior Packaging
MIL-F-13927	-	Fungus Resistance Test Automotive Components

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1275 (AT)	-	Characteristics of 28 Volt DC Electrical Systems in Military Vehicles
MIL-STD-2000	-	Standard Requirements for Soldered Electrical/Electronic Assemblies
MIL-STD-2073	-	DOD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12561737	-	VMS Modem Assembly
12561839	-	CCA2, VMS Modem
12561843	-	CCA1, VMS Modem

(Copies of drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form(s) a part of this document to the extent specified herein.

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Unless otherwise specified, the issue(s) of the document(s) which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-422 - Electrical Characteristics of Balanced Voltage Digital Interface Circuits

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, D.C. 20006):

2.3 Order of precedence. In the event of a conflict between the text of this document and the references herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 General description. The VMS modem operates by modulating discrete digital signals into frequency encoded signals and by demodulating frequency encoded signals into discrete digital signals.

3.2 Characteristics.

3.2.1 Performance. The VMS modem shall meet the performance requirements specified in the sections below.

3.2.1.1 J1 connector signal characteristics. All digital signal circuits shall be balanced differential voltage circuits in accordance with EIA standard RS-422. All line drivers shall comply with the generator electrical characteristics specified in EIA RS-422. All receivers shall comply with the requirements of EIA RS-422 with the following changes:

a. The common mode voltage range shall be -15 to +15 Vdc.

b. Over the entire common mode voltage range, the receiver shall not require a differential input voltage of more than 500 millivolts dc to correctly assume the intended binary state.

c. The maximum voltage (signal plus common mode) between either receiver input terminal and receiver circuit ground shall

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not exceed 18 Vdc in magnitude or cause the receiver circuit to operationally fail.

d. The allowable input current versus input voltage characteristics shall be extended as follows:

Input Voltage Range	Input Current Range
-18 to -10 Vdc	0 to -3.25 mA
+10 to +18 Vdc	0 to +3.25 mA

e. Over the entire common mode voltage range the balance of the receiver input voltage-current characteristics and bias voltages shall be such that the receiver will remain in the intended binary state when a differential voltage of 1000 millivolts is applied through 500 ohms plus or minus 1 percent to each input terminal.

The VMS modem shall not be damaged or malfunctioned when signal circuits are connected or disconnected with power on or off.

3.2.1.1.1 Logic levels. A binary "1" or assertion shall be represented by a positive voltage on the "Hi" line with respect to the "Lo" line. A binary "0" or negation shall be represented by a positive voltage on the "Lo" line with respect to the "Hi" line.

3.2.1.2 J2 connector signal characteristics. Data shall be transmitted and received via frequency-encoded signal circuits. Data will be formatted as a discrete frequency.

3.2.1.2.1 Logic levels. A high shall be represented by a voltage greater than 10.0 Vdc but less than 19.5 Vdc. A low shall be represented by a voltage less than 1.0 Vdc but greater than 0 Vdc.

3.2.1.2.2 Logic level output drive. The VMS modem shall be able to provide the output logic levels of 3.2.1.2.1 when driving a load that consists of a 5.0 ± 10 percent Vdc voltage source in series with 332 ± 10 percent ohms and a 1N4148 diode oriented so as to source current only, and also simultaneously driving no more than 0.018 microfarads shunted from the output to ground.

3.2.1.2.3 Logic level source impedance. The VMS modem shall respond to a source consisting of voltages of 18.0 ± 1.0 Vdc or 0.0 ± 0.5 Vdc in series with 33.2 ± 10 percent ohms and shunted by no more than 0.018 microfarads to ground producing the logic levels of 3.2.1.2.1.

3.2.1.3 Input power. The VMS modem normally operates from power supplied in accordance with MIL-STD-1275A(AT). The

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operational input voltage range for VMS modem power shall be 18.0 Vdc to 32.0 Vdc. This power shall be applied through J2 pin A with respect to return pin B.

3.2.1.3.1 Input current. The VMS modem shall draw 1.2 amps maximum at minimum input voltage of 18.0 Vdc and 1.2 amp maximum at maximum input voltage of 32.0 Vdc with the +5 Vdc pins (J1-J, K) tied together, the Digital Return pins (J1-R, S) tied together and a 10 ± 0.5 ohm resistor across the +5 Vdc and the Digital Return.

3.2.1.3.2 Power surge. The VMS modem shall survive a single voltage surge applied to the 18.0 to 32.0 Vdc inputs where the voltage increases from an initial value of 23.0 to 32.0 Vdc to its final value for a duration of 45.0 ± 5.0 milliseconds. The surge amplitude shall be 100V dc maximum. The surge voltage shall be provided by a supply with an output impedance not to exceed 500.0 milliohms. Surge rise and fall time shall be between 50.0 and 150.0 volts per millisecond. Surge voltage and rise and fall times shall be as stated from the supply before connection of the VMS modem.

3.2.1.3.3 Reverse power. The VMS modem shall survive simultaneous negative voltage inputs of up to -32.0 Vdc applied to the 18.0 to 32.0 Vdc inputs for a duration of 5.0 to 15.0 seconds. During the application of the reverse voltage, the VMS modem shall draw a total of less than 100.0 milliamperes from an external supply.

3.2.1.3.4 Transient protection. No damage shall occur to the VMS modem if the input voltage, as defined by MIL-STD-1275A (AT), has a voltage transient of ± 250 volts peak. This voltage transient shall be applied through J2 with pin connections as defined in 3.2.1.3.

3.2.1.4 VMS modem user interface. The VMS modem user interface shall include a power on indicator light. When power is applied in accordance with 3.2.1.3, the green power on light emitting diode (LED) shall be illuminated. When power is removed, the LED shall turn off.

3.2.1.5 VMS modem regulated power. The VMS modem shall be capable of providing a 5.0 ± 0.25 Vdc output when measured across R1 of Figure 1. Maximum output ripple shall be 300 mV (p-p) from DC to 20 MHz when a 10 µF capacitor is connected between the output and ground.

3.2.1.6 Internal loopback. The VMS modem, using the internal Built-in-Test (BIT) capability, shall generate pulse outputs according to the requirements of Table I when measured via EIA RS-422 compatible circuits as shown in Figure 1. The pulse shall

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be a single pulse which is 800 ± 24 µsec in width and shall be activated within 500 µs of the transition of J1-L, M or J1-N, P to a "1". Refer to Figure 2 for the internal loopback timing.

TABLE I. Internal loopback using J1.

Reqt	Inputs				Outputs 1/	
	J1-Z, a	J1-b, c	J1-L, M	J1-N, P	J1-V, W	J1-X, Y
1	0	1	0	0	0	0
2	0	1	0	1	0	Pulse
3	0	1	0	0	0	0
4	0	1	1	0	Pulse	0
5	0	1	0	0	0	0
6	0	1	1	1	Pulse	Pulse
7	0	1	0	0	0	0

1/ Transition from a 0 to a 1 causes a single pulse to be generated. Transition from a 1 to a 0 does not cause an output.

3.2.1.7 BIT command response. When J1-Z, a is set to 0 and J1-b,c is set to 1, the VMS modem shall be capable of setting J1-E, F; J1-G, H and J1-T,u output levels according to requirements 1 through 7 of Table II when measured via EIA RS-422 circuits as shown in Figure 1.

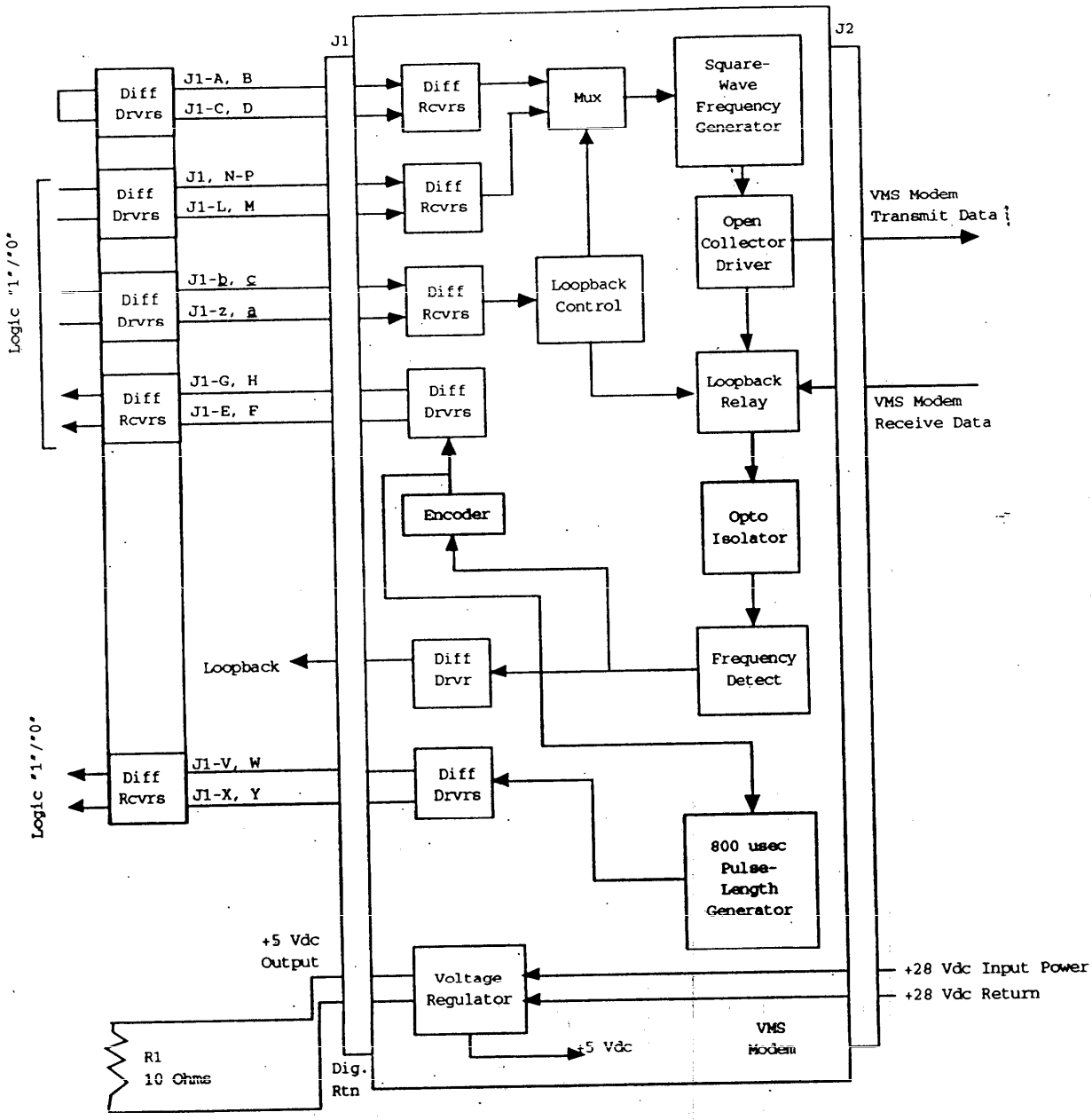
TABLE II. BIT command response.

Reqt	Input State		Output state transition			Delay (usec)	
	J1-L,M	J1-N,P	J1-E,F	J1-G,H	J1-T,U	Min	Max
1	0	0	0	0	0	no transitions	no transitions
2	1	0	0->1	0	n/a	100	300
3	0	0	1->0	0	n/a	n/a	50
4	0	1	0	0->1	n/a	50	300
5	1	1	0->1	1	n/a	100	500
6	0	1	1->0	1	n/a	n/a	100
7	0	0	0	1->0	n/a	n/a	50

Note: n/a is not applicable.

3.2.1.8 Transmit frequency capability. The VMS modem shall be capable of providing the steady state output frequencies within ± 3 percent accuracy which meet requirements 1 through 5 of Table III when measured at the VMS modem transmit data output (J2-C). The VMS modem shall also provide the BIT command 1 (J1-E, F), BIT command 2 (J1-G, H), and loopback outputs (J1-T, U) which meet requirements 1 through 5 of Table III with 6.25 kHz ± 3 percent applied to the VMS receive input (J2-D).

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Note: The differential receivers include a 120 ohm termination resistor.

FIGURE 1. VMS modem interface configuration (EIA RS-422 compatible circuit).

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TABLE III. VMS modem input/output requirements.

	Inputs							Outputs					
Req t	J1-C,D	J1-A,B	J1-Z,a	J1-b,c	J1-L,M	J1,N-P	Receive 6/ J2-D	J1-E,F	J1-G,H	J1-T,U	J1-V,W	J1-X,Y	J2-C
1	x	x	1	0	x	x	6.25 kHz	x	x	0	x	x	12.5 kHz
2	x	x	1	1	0	0	6.25 kHz	0	0	x	x	x	6.25 kHz
3	x	x	1	1	0	1	6.25 kHz	0	0	x	x	x	100 kHz
4	x	x	1	1	1	0	6.25 kHz	0	0	x	x	x	50 kHz
5	x	x	1	1	1	1	6.25 kHz	0	0	x	x	x	25 kHz
6	x	x	0	0	0	0	6.25 kHz	x	x	0	x	x	x
7	x	x	0	0	0	0	12.5 kHz 8/	x	x	1 1/	0	0	x
8	x	x	0	0	0	0	100 kHz 2/ 8/	x	x	0	0	1 3/	x
9	x	x	0	0	0	0	50 kHz 2/ 8/	x	x	0	1 3/	0	x
10	x	x	0	0	0	0	25 kHz 2/ 8/	x	x	0	1 3/	1 3/	x
11	Train 4/	0	0	0	x	x	x	1 5/	x	x	x	x	50 kHz 5/ 9/
12	0	Train 4/	0	0	x	x	x	x	1 5/	x	x	x	100 kHz 5/ 9/
13	1	1	0	0	x	x	x	x	x	x	x	x	25 kHz
14	0	0	0	0	x	x	x	x	x	x	x	x	6.25 kHz
	1/	1/			1/	1/	1/	1/	1/	1/	1/	1/	1/

1/ Output is as indicated for 0.1 sec. ± 10 percent. Delay for output to achieve indicated state from application of frequency on J2-D is 300 usec to 2 msec.
2/ VMS inputs are defined as follows:
pulse width (pw) = 800 ± 80 us.
pulse repetition frequency (prf) = 0 to 711 Hz ± 1 percent.
3/ Outputs are defined as follows:
pw = 800 ± 24 us
prf = same prf as input prf = ± 1 percent.
4/ "Train" inputs are defined as follows:
pw = 800 ± 80 us
prf = 0 to 711 Hz ± 1 percent.
5/ Outputs are defined as follows:
pw = same pw as input pw ± 5 percent
prf = same prf as input prf ± 1 percent.
6/ Frequency is ± 3 percent.
7/ X = Don't care
8/ During this test a 6.25 kHz signal shall be present at the J2-D input whenever the 12.5 kHz, 50 kHz, or 100 kHz signal is not being asserted (see figure 3).
9/ During this test a 6.25 kHz signal shall be present at the J2-C output whenever the 50 kHz or 100 kHz signal is not being asserted (see figure 3).

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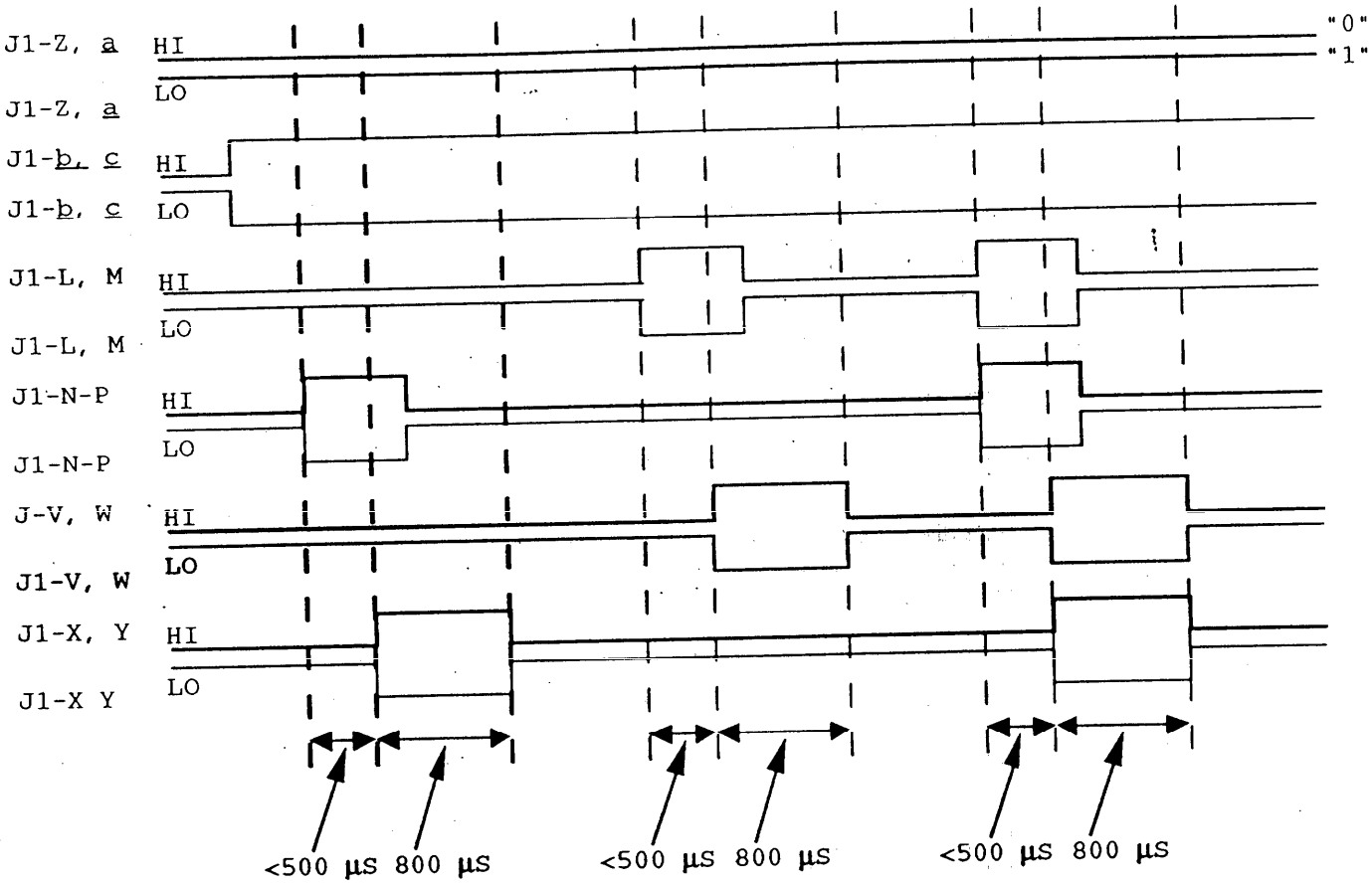


FIGURE 2. Timing waveform diagram for internal loopback.

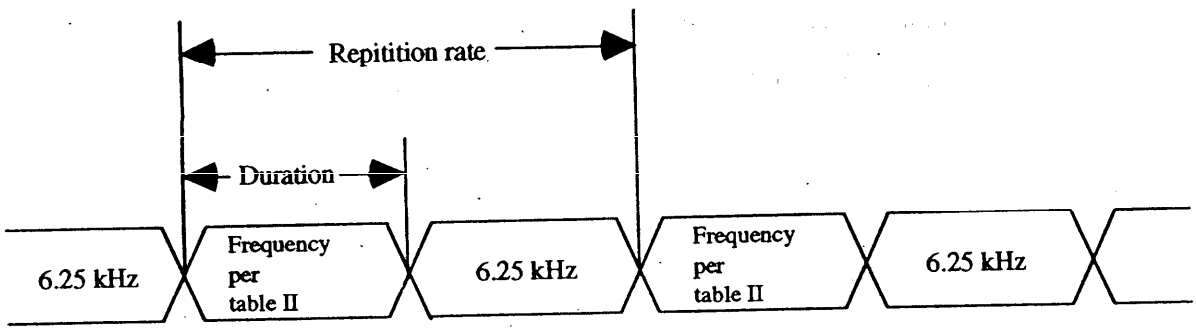


FIGURE 3. Timing waveform diagram for receive frequency capability and vehicle motion data transmit frequency.

3.2.1.9 Receive frequency capability. The VMS modem shall receive discrete frequency inputs (J2-D) and shall translate the inputs to the Odometer Forward, Odometer Reverse, and Loopback outputs in accordance with requirements 6 through 10 of Table III. For the frequency inputs of 25, 50, and 100 kHz, the VMS modem shall respond to input repetition rates from 0 through 711

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discrete frequency inputs per second with a response accuracy of ± 1 percent.

3.2.1.10 Vehicle motion data transmit frequency. The VMS modem shall receive Odometer Forward/BIT (J1-C, D) and Odometer Reverse/BIT 2 (J1-A, B) inputs and shall translate the inputs to outputs in accordance with requirements 11 through 14 of Table III. For the individual odometer forward and odometer reverse inputs, the VMS modem shall respond to an input repetition rate of 711 pulses per second with a response accuracy of ± 1 percent.

3.2.2 Environmental.

3.2.2.1 Operational environment.

3.2.2.1.1 Ambient operating temperature. The VMS modem shall meet the performance requirements in 3.2.1 while the operating temperature of the ambient air is within the range of -50°F to +140°F.

3.2.2.1.2 Humidity. The VMS modem shall meet the performance requirements in 3.2.1 at a relative humidity up to a maximum of 100 percent across the temperature range defined in 3.2.2.1.1.

3.2.2.1.3 Salt fog. The VMS modem shall meet the performance requirements in 3.2.1 during and after exposure to salt fog solution at a temperature between 90°F and 95°F. The solution is defined as 5 percent Sodium Chloride and 95 percent distilled water by weight. Fog density is approximately 1.057 quarts of salt solution per 264 gallons of chamber volume per 24 hours.

3.2.2.1.4 Vibration. The VMS modem shall meet the performance requirements in 3.2.1 during and after being subjected to vibration as shown in Table IV using the axis configurations as shown in Figure 4.

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TABLE IV. Vibration.

Test Phase	5-500 Hz Floor Level (g^2/Hz)	No. Sweeps	Narrowband 1 BW (Hz)	Narrowband 1 Ampl (g^2/Hz)	Narrowband 1 Sweep BW (Hz)	Narrowband 2 BW (Hz)	Narrowband 2 Ampl (g^2/Hz)	Narrowband 2 Sweep BW (Hz)	Narrowband 3 BW (Hz)	Narrowband 3 Ampl (g^2/Hz)	Narrowband 3 Sweep BW (Hz)	Narrowband 4 BW (Hz)	Narrowband 4 Ampl (g^2/Hz)	Narrowband 4 Sweep BW (Hz)	Narrowband 5 BW (Hz)	Narrowband 5 Ampl (g^2/Hz)	Narrowband 5 Sweep BW (Hz)
						VERTICAL AXIS (11 minutes per test phase)											
V01	0.0010	6	18-30	0.0882	6	36-60	0.0081	12	-------	------	--	-------	------	--	-------	------	
V02	0.0024	6	36-42	0.4914	3	72-84	0.0391	6	108-126	0.0076	9	144-168	0.0275	12	-------	------	--
V03	0.0026	2	48-60	0.2467	6	96-120	0.0319	12	144-180	0.0062	18	192-240	0.0135	24	-------	------	--
V04	0.0034	2	66-96	0.1227	15	132-192	0.0249	30	198-288	0.0074	45	-------	------	--	-------	------	--
V05	0.0024	2	102-126	0.0307	12	204-252	0.0120	24	408-500	0.0079	48	-------	------	--	-------	------	--
						TRANSVERSE AXIS (11 minutes per test phase)											
T01	0.0022	4	18-36	0.0585	9	36-72	0.0064	18	-------	------	--	-------	------	--	-------	------	
T02	0.0034	4	42-48	0.1639	3	84-96	0.0435	6	126-144	0.0128	9	168-192	0.0130	12	210-240	0.0239	15
T03	0.0032	2	54-66	0.0332	6	108-132	0.0223	12	162-198	0.0150	18	216-264	0.0179	24	270-330	0.0152	30
T04	0.0040	2	72-96	0.0457	12	144-192	0.0221	24	216-288	0.0660	36	288-384	0.0076	48	-------	------	--
T05	0.0030	2	102-126	0.0024	12	204-252	0.0647	24	306-378	0.0072	36	408-500	0.0074	48	-------	------	--
						LONGITUDINAL AXIS (11 minutes per test phase)											
L01	0.0012	4	18-36	0.0069	9	36-72	0.0046	18	-------	------	--	-------	------	--	-------	------	
L02	0.0017	4	42-48	0.0305	3	84-96	0.0143	6	126-144	0.0040	9	168-192	0.0246	12	210-240	0.0037	15
L03	0.0022	2	54-66	0.0679	6	108-132	0.0034	12	162-198	0.0067	18	216-264	0.0032	24	-------	------	--
L04	0.0030	2	72-90	0.0369	9	144-180	0.0229	18	216-270	0.0040	27	288-360	0.0052	36	360-450	0.0069	45
L05	0.0022	2	96-126	0.0157	15	192-252	0.0091	30	288-378	0.0076	45	384-500	0.0072	60	-------	------	--

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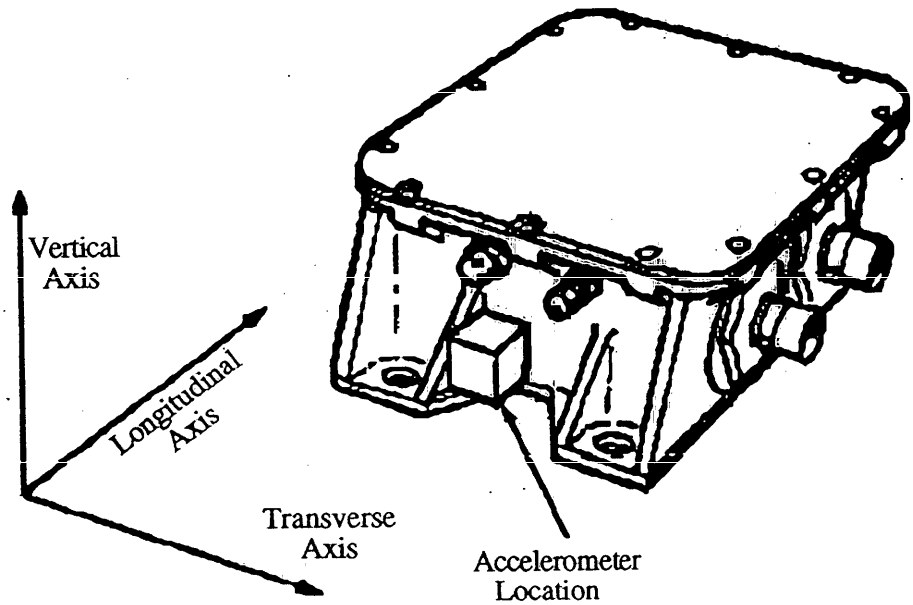


FIGURE 4. VMS modem axes orientations.

3.2.2.1.5 Shock.

3.2.2.1.5.1 Basic shock. The VMS modem shall meet the performance requirements in 3.2.1 after being subjected to three half sine wave pulses applied along each of the three mutually perpendicular axes shown in Figure 4 with peak amplitudes of 30 ± 3 g for 18 ± 1.8 msec.

3.2.2.1.5.2 Gun firing shock. The VMS modem shall meet the performance requirements in 3.2.1 during and after being subjected to three half sine wave pulses applied along each of three mutually perpendicular axes with peak amplitudes of 200 ± 20 g for 0.5 ± 0.1 msec as shown in Figure 4.

3.2.2.1.6 Altitude. The VMS modem shall meet the performance requirements in 3.2.1 during and after exposure to pressure encountered at altitudes up to 10,000 feet above sea level (10.1 psia).

3.2.2.2 Nonoperational environment.

3.2.2.2.1 Ambient storage temperature. The VMS modem shall meet the performance requirements in 3.2.1 after being subject to storage temperature of the ambient air within the range of -50°F to +160°F per the following restrictions:

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a. Exposure at temperature extremes of 145°F to 160°F shall be limited to one hour duration per 24-hour period.

b. Temperature variation during any 8-hour period shall be sinusoidal with a maximum variation of 69°F.

3.2.2.2.2 Fungus. The VMS modem shall meet the performance requirements in 3.2.1 after exposure to spore suspension as defined in the "Specimen Inoculation" paragraph of MIL-F-13927, followed by exposure to ambient air temperatures between 81°F and 93°F at relative humidity between 96 and 100 percent, noncondensing for a 28 day duration. All exterior connectors shall have a mating connector or equivalent installed during exposure.

3.2.2.2.3 Altitude. The VMS modem shall meet the performance requirements in 3.2.1 after exposure to pressures within the range of 1.68 to 2.14 psia.

3.2.2.2.4 Washdown. The VMS modem shall meet the performance requirements in 3.2.1 after being subjected to washdown using hot water at 140°F ± 20°F with a operating (tank or reservoir) pressure between 9.0 and 11.0 psi and air drying at a temperature not to exceed 145°F. All exterior connectors shall have a mating cable, connector cover, or equivalent installed prior to exposure.

3.3 Environmental Stress Screening (ESS). Unless otherwise specified, ESS shall be performed as specified below (see 6.2).

3.3.1 SRU ESS. Unless otherwise specified, the following SRUs shall be subjected to environmental stress screening as specified below (see 6.2 herein):

Part number	Nomenclature
12561843	VMS Modem 1
12561839	VMS Modem 2

3.3.1.1 Environments. The SRUs shall be subjected first to vibration, then to temperature cycling. The SRUs shall not have power applied during either the vibration or temperature portions of the ESS.

3.3.1.1.1 Vibration. The SRUs shall be subject to random vibration per Figure 5 in each of the three axes for 10 minutes per axis. The average vibration level shall be monitored at the SRUs mounting points and shall be controlled using a control bandwidth of no greater than 10 Hz for vibration frequencies above 500 Hz. Instantaneous acceleration peaks shall be limited to three times the Root-Mean Square acceleration value.

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3.3.1.1.2 Temperature cycling. The SRUs shall be subjected to 20 cycles, per Figure 6 with the following exceptions: maximum temperature shall be 167 °F and the minimum temperature shall be -58 °F. Chamber air temperature changes shall be at the rate of 18°F/minute. Each temperature extreme shall be maintained until the SRUs temperature change is less than 9°F/hour.

3.3.1.1.3 Post cycle test. The SRUs shall be checked with the following performance tests following exposure to ESS environments.

a. The VMS Modem 1 CCA shall, at a minimum, perform a functional test which checks the following functions:

- Transmit frequencies
- BIT command
- Loopback/odometer pulse width
- Loopback relay

b. The VMS Modem 2 CCA shall, at a minimum, perform a functional test which checks the following functions:

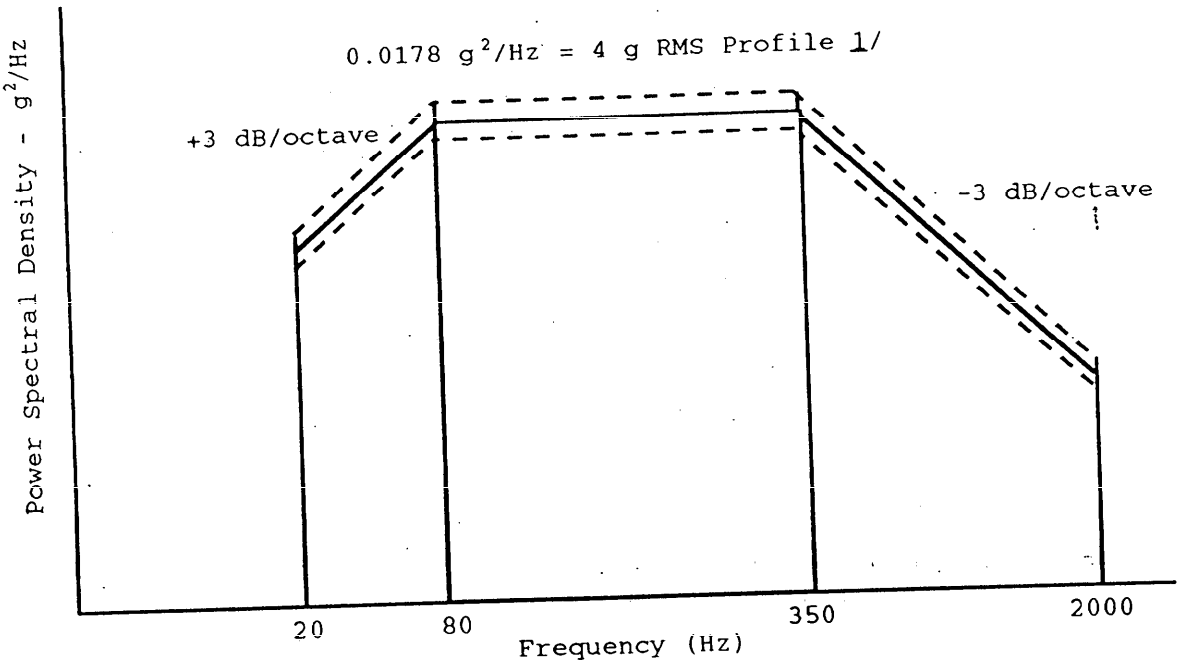
- Power Output
- Frequency Detects

3.3.2 LRU ESS. The VMS modem shall be subjected to environmental stress screening.

3.3.2.1 Environments. The VMS modem shall be subjected first to vibration, then to temperature cycling.

3.3.2.1.1 Vibration. The VMS modem shall be subjected to random vibration per Figure 5 in each of the three axes for 10 minutes per axis. The average vibration level shall be monitored at the VMS modem mounting points and shall be controlled using a control bandwidth of no greater than 10 Hz for vibration frequencies above 500 Hz. Instantaneous acceleration peaks shall be limited to three times the Root-Mean-Square acceleration value.

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1/ Dashed line indicates tolerance band of ± 3 dB.

FIGURE 5. Random vibration spectrum.

3.3.2.1.2 Temperature cycling. The VMS modem shall be subjected to twelve temperature cycles per Figure 6. Chamber air temperature changes shall be at the rate of 18°F/minute. Each temperature extreme shall be maintained until the VMS modem temperature change is less than 9°F/hour.

3.3.2.2 Monitoring. The VMS modem shall have power applied to it, and be monitored, during vibration and portions of the temperature cycles per Figure 6. If a failure occurs during vibration, vibration shall be repeated after LRU repair. If a failure occurs during any except the last three temperature cycles, temperature cycling shall be restarted, after LRU repair, at the beginning of the cycle on which the VMS modem failed. If a failure occurs during the last three temperature cycles, temperature cycling shall be restarted, after LRU repair, at the beginning of the third from the last cycle. If the VMS modem is subjected to vibration three times or accumulates twelve temperature cycles before completing ESS, it shall be subjected to formal failure analysis. The following parameters shall be repetitively verified:

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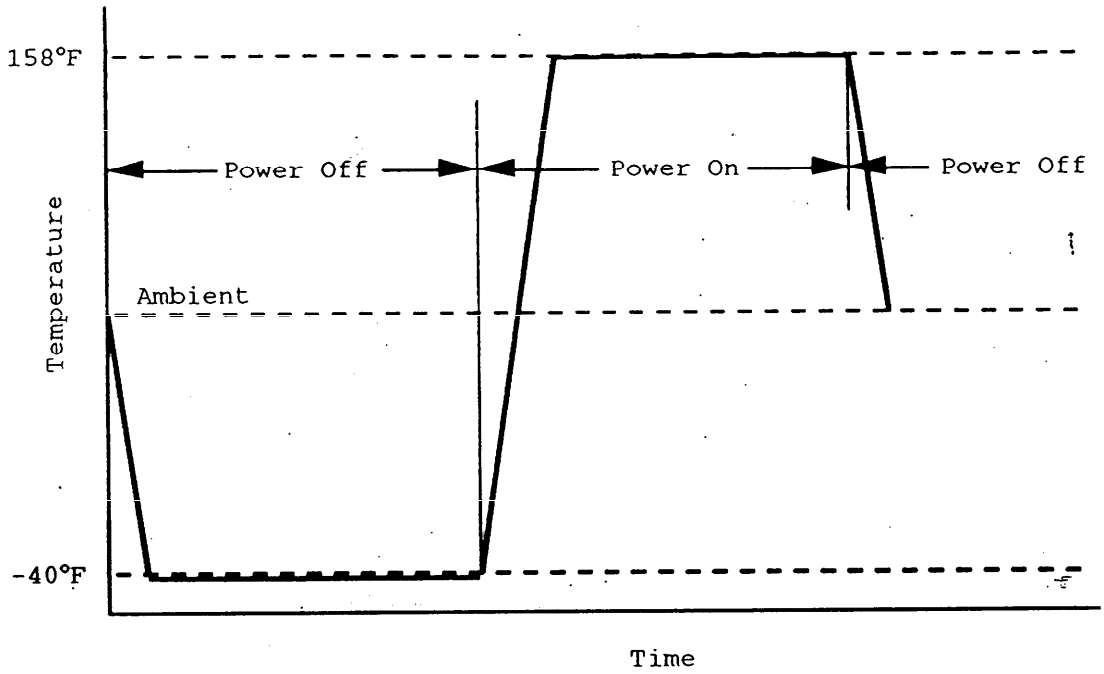


FIGURE 6. Temperature cycling profile.

Parameter	Verification
Internal loopback, forward	3.2.1.6
Internal loopback, reverse	3.2.1.6

3.4 Fabrication. The VMS modem shall be manufactured in accordance with drawing 12561737.

3.5 First article When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.5.

3.5.1 First article sample inspection. After completion of first article test, the first article sample shall meet the requirements of 3.6.1.

3.6 Workmanship.

3.6.1 General workmanship. General workmanship shall be in accordance with MIL-STD-454, Requirement 9.

3.6.2 Solder workmanship. All soldering shall be in accordance with MIL-STD-2000 (See 6.2).

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.5).
b. Quality Conformance (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions in 4.7.1.

4.4 Inspection equipment. Contractor developed special test equipment including test software required to perform examinations and tests prescribed herein shall be submitted for approval in accordance with the terms of the contract (see 6.2).

4.5 First article inspection.

4.5.1 Submission. When required by the procuring activity (see 6.2), the contractor shall submit a First Article sample for evaluation in accordance with the provisions of 4.5.2 and 4.5.3. The First Article sample size shall be three (3) or as specified in the contract or purchase order.

4.5.2 Inspections to be performed. The sequence of first article testing shall be in accordance with Table V. The first article sample shall have been subjected to, and pass, the Quality Conformance Requirements as defined in Table VI. Subassemblies

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and piece parts, at a minimum, shall be inspected for conformance to the characteristics defined in the quality assurance provisions of the applicable drawing.

TABLE V. First article test sequence.

Test Sequence 1/	Requirement Paragraph	Test Method	Sample 1	2	3
Performance 2/	3.2.1	4.7.2	X	X	X
Cold ambient storage temperature	3.2.2.2.1	4.7.3.2.1.1	X	X	X
Hot ambient storage temperature	3.2.2.2.1	4.7.3.2.1.2	X	X	X
Cold ambient operating temperature	3.2.2.1.1	4.7.3.1.1.1	X	X	X
Hot ambient operating temperature	3.2.2.1.1	4.7.3.1.1.2	X	X	X
Altitude nonoperating	3.2.2.2.3	4.7.3.2.3	X	X	X
Altitude operating	3.2.2.1.6	4.7.3.1.6	X	X	X
Vibration	3.2.2.1.4	4.7.3.1.4	X	X	X
Basic shock	3.2.2.1.5.1	4.7.3.1.5.1	X	X	X
Gun firing shock	3.2.2.1.5.2	4.7.3.1.5.2	X	X	X
Washdown	3.2.2.2.4	4.7.3.2.4	X	X	X
Humidity	3.2.2.1.2	4.7.3.1.2	X		
Fungus	3.2.2.2.2	4.7.3.2.2		X	
Salt fog	3.2.2.1.3	4.7.3.1.3			X
First article sample inspection	3.5.1	4.7.3.3	X	X	X

1/ To ensure the first article sample has been subjected to, and passed, the quality conformance tests as defined in Table VI the contractors records may be reviewed to assess compliance.

2/ The first article performance tests, as defined in Table V, shall be performed only once. Performance testing, after the first article sample has been subjected to the various environments shall consist of the performance test defined in the quality conformance section of Table VI.

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TABLE VI. Category of inspection.

Class.	Description	Requirements	Method 1/	First Article	Quality Conformance
	Performance	3.2.1			
	Input power	3.2.1.3	4.7.2.1	X	100%
Minor	Input current	3.2.1.3.1	4.7.2.1.1	X	100%
Minor	Power surge	3.2.1.3.2	4.7.2.1.2	X	
Minor	Reverse power	3.2.1.3.3	4.7.2.1.3	X	
Minor	Transient protection	3.2.1.3.4	4.7.2.1.4	X	
Major	VMS modem user interface	3.2.1.4	4.7.2.2	X	100%
Major	VMS modem regulated power	3.2.1.5	4.7.2.3	X	100%
Major	Regulated power ripple	3.2.1.5	4.7.2.3.1	X	100%
Major	Internal loopback	3.2.1.6	4.7.2.4	X	100%
Major	Bit command response	3.2.1.7	4.7.2.5	X	100%
Major	Transmit frequency capability	3.2.1.8	4.7.2.6	X	100%
Major	Receive frequency capability	3.2.1.9	4.7.2.7	X	100%
Major	Vehicle motion data transmit frequency	3.2.1.10	4.7.2.8	X	100%
Major	Environmental	3.2.2		X	
Major	Operational environment	3.2.2.1	4.7.3.1	X	
Major	Ambient operating temperatures	3.2.2.1.1	4.7.3.1.1	X	
Major	Cold temperature	3.2.2.1.1	4.7.3.1.1.1	X	
Major	Hot temperature	3.2.2.1.1	4.7.3.1.1.2	X	
Major	Humidity	3.2.2.1.2	4.7.3.1.2	X	
Major	Salt fog	3.2.2.1.3	4.7.3.1.3	X	
Major	Vibration	3.2.2.1.4	4.7.3.1.4	X	
Major	Shock	3.2.2.1.5	4.7.3.1.5		
Major	Basic shock	3.2.2.1.6.1	4.7.3.1.5.1	X	
Major	Gun firing shock	3.2.2.1.5.2	4.7.3.1.5.2	X	
Major	Altitude	3.2.2.1.6	4.7.3.1.6	X	
	Nonoperational Environments	3.2.2.2	4.7.3.2	X	
	Ambient storage Temperature	3.2.2.2.1	4.7.3.7.1	X	
Major	Cold temperature	3.2.2.2.1	4.7.3.2.1.1	X	
Major	Hot temperature	3.2.2.2.1	4.7.3.2.1.2	X	
Major	Fungus	3.2.2.2.2	4.7.3.2.2	X	
Major	Altitude	3.2.2.2.3	4.7.3.2.3	X	
Major	Washdown	3.2.2.2.4	4.7.3.2.4	X	
Minor	First Article Sample Insp.	3.5.1	4.7.3.3	X	
Major	Environmental stress screening	3.3	4.7.4	X	100%
	Workmanship	3.6	4.7.5		X
Minor	General workmanship	3.6.1	4.7.5.1	X	2/
Minor	Solder workmanship	3.6.2	4.7.5.2	X	100%
Minor	Packaging	5	4.7.6		2/

1/ Unless otherwise specified in 4.6, sequence of functional tests and physical characteristic examinations are optional.

2/ Sample size is as defined by level II of Table VII.

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4.5.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements. Disposition of the item shall be as specified in the contract or purchase order (see 6.3).

4.6 Quality conformance inspections.

4.6.1 Examinations and tests.

4.6.1.1 Classification of characteristics. Quality conformance examination and test shall be as specified in Table VI. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Subassemblies and piece parts shall be inspected for conformance to the characteristics defined in the quality assurance provisions of the applicable drawing.

TABLE VII. Attributes sampling inspection.

Inspection levels 1/		
Lot Size	I	II
2 to 8	*	*
9 to 15	*	13
16 to 25	*	13
26 to 50	*	13
51 to 90	*	13
91 to 150	125	13
151 to 280	125	32

1/ Numbers under inspection levels indicate sample size: asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

4.6.1.2 Alternative quality conformance provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe

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them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.6.2 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the item shall be rejected. Disposition of the item shall be as specified in the contract or purchase order. (See 6.2.)

4.7 Methods of inspection.

4.7.1 Test conditions.

4.7.1.1 Environmental. Unless otherwise specified, all tests shall be performed at an atmospheric pressure between 28 and 32 inches of mercury, a temperature of 75° ± 25°F, and a relative humidity between 10 percent and 90 percent.

4.7.1.2 Test software. Test software includes supporting test software which resides in external test computer(s). Test software is required for testing the performance requirements of (4.7.2). Test software used for first article and quality conformance tests shall be submitted for approval (see 4.4) as part of the inspection test equipment.

4.7.1.3 Input power. All tests shall be performed at a nominal input voltage of 28 ± 0.25 Vdc and power shall be applied to the VMS modem as defined in 3.2.1.3, unless otherwise specified.

4.7.1.4 Input logic levels.

4.7.1.4.1 J1 logic levels. Whenever J1 input levels are to be provided to the unit under test they shall meet the requirements of EIA RS-422 as described in section 3.2.1.1.1.

4.7.1.4.2 J2 logic levels. The logic levels of J2 connector shall meet the requirements of 3.2.1.2.1.

4.7.2 Performance test methods.

4.7.2.1 Input power.

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4.7.2.1.1 Input current. The input current shall not exceed the maximum values specified in Table VIII, when tested at the minimum input voltage (Requirement 1) and the maximum input voltage (Requirement 2). This test shall be performed with input power applied through connector J2 and a load applied to the VMS modem +5 Vdc output as defined in 3.2.1.3.1.

TABLE VIII. Input power requirement.

Requirement	Input Voltage Volts Vdc	Input Current Amps (max)
1	18 +0.5/-0.0	1.2
2	32 +0.0/-0.5	1.2

4.7.2.1.2 Power surge. The VMS modem shall meet the performance requirements of s 4.7.2.2 through 4.7.2.8 after being subjected to a 100 +0/-10 Vdc power surge as specified in 3.2.1.3.2.

4.7.2.1.3 Reverse power. The VMS modem shall meet the performance requirements of 4.7.2.2 through 4.7.2.8 after being subjected to the negative input voltage specified in 3.2.1.3.3 for a period of 10 ± 5 seconds. The current draw from the external voltage supply shall not exceed the requirements specified in 3.2.1.3.3.

4.7.2.1.4 Transient protection. The VMS modem shall meet the performance requirements of 4.7.2.2 through 4.7.2.8 after being subjected to a transient as specified in 3.2.1.3.4.

4.7.2.2 VMS modem user interface. When power is applied to the VMS modem in accordance with 4.7.1.3, the green power light illumination shall be visually verified. When power is removed, the green power light shall be turned off.

4.7.2.3 VMS modem regulated power. With power applied to the VMS modem in accordance with 4.7.1.3, measure the VMS modem regulated power output voltage across resistor R1 of Figure 1. VMS modem regulated power output voltage shall be in accordance with the requirements of 3.2.1.5.

4.7.2.3.1 Regulated power ripple. Measure the ripple voltage on the regulated power output. The ripple voltage shall not exceed the requirements of 3.2.1.5.

4.7.2.4 Internal loopback. The internal loopback function of the VMS modem shall be tested as follows to verify the requirements of 3.2.1.6:

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a. Set Test 1 (J1-Z,a) to "0" and Test 2 (J1-b,c) to "1".

b. (Requirement 1) Set CU BIT 1 (J1-L,M) to "0" and CU BIT 2 (J1-N,P) to "0" and verify that Odometer Forward (J1-V,W) is "0" and Odometer Reverse (J1-X,Y) is "0".

c. (Requirement 2) Set CU BIT 1 to "0" and CU BIT 2 to "1" and verify that Odometer Forward is "0" and Odometer Reverse outputs a pulse of width 800 ± 24 µsec within 500 µsec of CU BIT 2 transitioning to a "1".

d. (Requirement 3) Set CU BIT 1 to "0" and CU BIT 2 to "0" and verify that Odometer Forward is "0" and Odometer Reverse is "0".

e. (Requirement 4) Set CU BIT 1 to "1" and CU BIT 2 to "0" and verify that Odometer Reverse is "0" and Odometer Forward outputs a pulse of width 800 ± 24 µsec within 500 µsec of CU BIT 1 transitioning to a "1".

f. (Requirement 5) Set CU BIT 1 to "0" and CU BIT 2 to "0" and verify that Odometer Forward is "0" and Odometer Reverse is "0".

g. (Requirement 6) Set CU BIT 1 to "1" and CU BIT 2 to "1" and verify that both Odometer Reverse and Odometer Forward output a pulse of width 800 ± 24 µsec within 500 µsec of CU BIT 1 and CU BIT 2 transitioning to a "1".

h. (Requirement 7) Set CU BIT 1 to "0" and CU BIT 2 to "0" and verify that Odometer Forward is "0" and Odometer Reverse is "0".

4.7.2.5 BIT command response. The BIT command response function of the VMS modem shall be tested as follows to verify the requirements of 3.2.1.7. A minimum of 20 measurements shall be made for each delay measurement for requirements 2 through 7. The minimum and maximum readings from those 20 measurements shall be used to determine the minimum and maximum delay.

a. Set Test 1 (J1-Z,a) to "0" and Test 2 (J1-b,c) to "1".

b. (Requirement 1) Set CU BIT 1 (J1-L,M) to "0" and CU BIT 2 (J1-N,P) to "0" and verify that BIT Command 1 (J1-E,F) is "0" and BIT Command 2 (J1-G,H) is "0". Verify that Loopback (J1-T,U) is "0".

c. (Requirement 2) Set CU BIT 1 to "1" and CU BIT 2 to "0" and verify the output transition and delay of Table II.

d. (Requirement 3) Set CU BIT 1 to "0" and CU BIT 2 to "0" and verify the output transition and delay of Table II.

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e. (Requirement 4) Set CU BIT 1 to "0" and CU BIT 2 to "1" and verify the output transition and delay of Table II.

f. (Requirement 5) Set CU BIT 1 to "1" and CU BIT 2 to "1" and verify the output transition and delay of Table II.

g. (Requirement 6) Set CU BIT 1 to "0" and CU BIT 2 to "1" and verify the output transition and delay of Table II.

h. (Requirement 7) Set CU BIT 1 to "0" and CU BIT 2 to "0" and verify the output transition and delay of Table II.

4.7.2.6 Transmit frequency capability. The transmit frequency function of the VMS modem shall be tested as follows to verify the requirements of 3.2.1.8:

a. Apply a 6.25 kHz ± 3 percent signal to the VMS Receive (J2-D) input for the entire test.

b. (Requirement 1) Set Test 1 (J1-Z,a) to "1" and Test 2 (J1-b,c) to "0" and verify a continuous 12.5 kHz ± 3 percent signal on the VMS Transmit (J2-C) output. Also verify Loopback (J1-T,U) is "0".

c. Set both Test 1 and Test 2 inputs to "1".

d. (Requirement 2) Set CU BIT 1 (J1-L,M) to "0" and CU BIT 2 (J1-N,P) to "0" and verify a continuous 6.25 kHz ± 3 percent signal on the VMS Transmit output. Also verify BIT Command 1 (J1-E,F) and BIT Command 2 (J1-G,H) outputs are "0".

e. (Requirement 3) Set CU BIT 1 to "0" and CU BIT 2 to "1" and verify a continuous 100 kHz ± 3 percent signal on the VMS Transmit output. Also verify BIT Command 1 and BIT Command 2 outputs remain "0".

f. (Requirement 4) Set CU BIT 1 to "1" and CU BIT 2 to "0" and verify a continuous 50 kHz ± 3 percent signal on the VMS Transmit output. Also verify BIT Command 1 and BIT Command 2 outputs remain "0".

g. (Requirement 5) Set CU BIT 1 to "1" and CU BIT 2 to "1" and verify a continuous 25 kHz ± 3 percent signal on the VMS Transmit output. Also verify BIT Command 1 and BIT Command 2 outputs remain "0".

4.7.2.7 Receive frequency capability. The receive frequency function of the VMS modem shall be tested as follows to verify the requirements of 3.2.1.9:

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a. Set Test 1 (J1-Z,a), Test 2 (J1-b,c), CU BIT 1 (J1-L,M), and CU BIT 2 (J1-N,P) all to "0" for the duration of the test.

b. (Requirement 6) Apply a 6.25 kHz ± 3 percent signal to the VMS Receive input. Verify Odometer Forward (J1-V,W) and Odometer Reverse (J1-X,Y) are both "0" and Loopback is a "0".

c. (Requirement 7) Apply a 12.5 kHz ± 3 percent signal in accordance with Table III for a minimum duration of 2 msec to the VMS Receive (J2-D) input and verify Loopback (J1-T,U) is a "1" for 0.1 sec ± 10 percent. Verify the loopback delay is between 300 µsec and 2 msec.

d. (Requirement 8) Apply a 100 kHz ± 3 percent signal in accordance with Table III for a duration of 880 +0/-10 µsec to the VMS Receive input at repetition rates of 1, 356, and 711 Hz. Verify that Odometer Reverse outputs a pulse of 800 ± 24 µsec at the same repetition rate as the input ± 1 percent. Verify Odometer Forward is a "0" and Loopback is a "0".

e. (Requirement 9) Apply a 50 kHz ± 3 percent signal in accordance with Table III for a duration of 720 +10/-0 µsec to the VMS Receive input at repetition rates of 1, 356, and 711 Hz. Verify that Odometer Forward outputs a pulse of 800 ± 24 µsec at the same repetition rate as the input ± 1 percent. Verify Odometer Reverse is a "0" and Loopback is a "0".

f. (Requirement 10) Apply a 25 kHz ± 3 percent signal in accordance with Table III for a duration of 880 +0/-10 µsec to the VMS Receive input at a repetition rate of 711 Hz. Verify that both Odometer Forward and Odometer Reverse output a pulse of 800 ± 24 µsec at the same repetition rate as the input ± 1 percent. Verify Loopback is a "0".

4.7.2.8 Vehicle motion data transmit frequency. The vehicle motion data transmit frequency function of the VMS modem shall be tested as follows to verify the requirements of 3.2.1.10:

a. Set Test 1 (J1-Z,a) and Test 2 (J1-b,c) to "0" for the duration of the test.

b. (Requirement 11) Apply a pulse train input with pulses of 880 +0/-10 µsec to Odometer Forward/BIT 1 (J1-C,D) at a repetition rate of 711 Hz. Verify that VMS transmit (J2-C) outputs a 50 kHz ± 10 percent signal with the same duration as the input pulse width ± 5 percent and the same repetition rate as the input ± 1 percent.

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c. (Requirement 12) Apply a pulse train input with pulses of 720 +10/-0 µsec to Odometer Reverse/BIT 2 (J1-A,B) at a repetition rate of 711 Hz. Verify that VMS transmit outputs a 100 kHz ± 10 percent signal with the same duration as the input pulse width ± 5 percent and the same repetition rate as the input ± 1 percent.

d. (Requirement 13) Set Odometer Forward/BIT 1 and Odometer Reverse/BIT 2 both to "1". Verify that VMS transmit outputs a 25 kHz ± 10 percent signal.

e. (Requirement 14) Set Odometer Forward/BIT 1 and Odometer Reverse/BIT 2 both to "0". Verify that VMS transmit outputs a 6.25 kHz ± 10 percent signal.

4.7.3 Environmental test methods.

4.7.3.1 Operational environment.

4.7.3.1.1 Ambient operating temperature.

4.7.3.1.1.1 Cold ambient operating temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 502.2, Procedure II after thermally stabilizing at a low temperature of -50° +5°/-0°F for a minimum of 4 hours. During the stabilization period the unit shall be unpowered. At the conclusion of this time, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI. If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the temperature conditions shall be re-established prior to continuing.

4.7.3.1.1.2 Hot ambient operating temperature. This test shall be conducted in accordance with and MIL-STD-810D, Method 501.2, Procedure II after thermally stabilizing at a high temperature of +140° +0°/-5°F for a minimum of 4 hours. During the stabilization period the unit shall be powered. At the conclusion of this time, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI. If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the temperature conditions shall be re-established prior to continuing.

4.7.3.1.2 Humidity. This test shall be conducted in accordance with MIL-STD-810D, Method 507.2, Procedure I, cycle 2, for 240 hours (10 cycles) except that the temperature shall be cycled from 85°F to 140°F during each 24 hour cycle. During each 24 hour cycle the temperature will be stabilized at 85°F for a

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minimum of 4 hours and at 140°F for a minimum of 4 hours. The rate of temperature change during transition periods shall not exceed 8°F/hour. All exterior connectors shall have a mating connector, cover, or equivalent installed prior to exposure. Operational checkouts shall be conducted using the quality conformance performance tests specified in Table VI and shall be performed at the end of the fifth and tenth cycles as a minimum. If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the test conditions shall be re-established prior to continuing.

4.7.3.1.3 Salt fog. This test shall be conducted in accordance with MIL-STD-810D, Method 509.2. The unit under test shall be unpowered while exposed to a salt fog solution of 5 percent ± 1 percent sodium chloride for 48 hours. The temperature in the exposure zone shall be maintained between 90°F and 95°F. All exterior connectors shall have a mating connector cover or equivalent installed prior to exposure. After exposure, the VMS modem shall be dried for 48 hours and then tested per the quality conformance performance tests specified in Table VI.

4.7.3.1.4 Vibration. This test shall be conducted in accordance with MIL-STD-810D, Method 514.3, Procedure I, category 8. The vibration test levels and duration are specified in Table IV in 3.2.2.1.4. After exposure, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI. Location of the vibration control accelerometers shall be as shown in Figure 4. Test duration per phase shall be as specified in Table III plus or minus 30 seconds.

4.7.3.1.5 Shock.

4.7.3.1.5.1 Basic shock. This test shall be conducted in accordance with MIL-STD-810D, Method 516.3, Procedure I and shall consist of 3 cycles of shock as shown in 3.2.2.1.5.1. The response spectra shall not deviate from the input condition by more than ± 3.5 dB. After exposure, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI.

4.7.3.1.5.2 Gunfire shock. This test shall be conducted in accordance with MIL-STD-810D, Method 516.3, Procedure I and shall consist of 3 cycles of shock as shown in 3.2.2.1.5.2. The response spectra shall not deviate from the input condition by more than ± 3.5 dB. After exposure, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI.

4.7.3.1.6 Altitude. This test shall be conducted in accordance with MIL-STD-810D, Method 500.2, Procedure II. Pressure shall be maintained at 10.1 psia for a minimum of 30

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minutes. During the stabilization period the unit shall be unpowered. At the conclusion of this time VMS modem shall be tested per the quality conformance performance test specified in Table VI.

4.7.3.2 Nonoperational environment.

4.7.3.2.1 Ambient storage temperature.

4.7.3.2.1.1 Low ambient storage temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 502.2, Procedure I for 168 hours at a low temperature of -50° +5°/-0°F. After exposure, the VMS modem shall be tested per the quality performance tests specified in Table VI at -50° +5°/-0°F and again at room temperature (75° ± 25°F). If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the temperature conditions shall be re-established prior to continuing.

4.7.3.2.1.2 High ambient storage temperature. This test shall be conducted in accordance with MIL-STD-810D, Method 501.2, Procedure I for 168 hours (7 cycles) at a high temperature of +145° +0°/-5°F, with 1 hour +0/-5 minutes of each 24 hour cycle at +160° +0°/-5°F per Figure 7. After exposure, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI at 140° +0°/-5°F and again at room temperature (75° ± 25°F). If needed, the chamber door may be opened to take measurements directly or the unit may be removed from the chamber to perform the tests. These tests shall be completed within 15 minutes of opening the door, if not completed within 15 minutes, the temperature conditions shall be re-established prior to continuing.

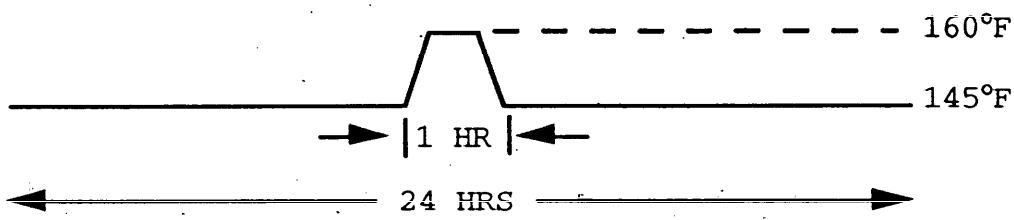


FIGURE 7. Hot storage temperature profile.

4.7.3.2.2 Fungus. This test shall be conducted in accordance with MIL-F-13927, Class 2, Method B except temperature and duration shall be as specified in 3.2.2.2.2. All exterior connectors shall have a mating connector cover or equivalent installed prior to exposure. After exposure, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI at room temperature (75° ± 25°F).

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4.7.3.2.3 Altitude. This test shall be conducted in accordance with MIL-STD-810D, Method 500.2, Procedure I and 3.2.2.2.3 pressure shall be maintained at a specified value for 4 hours -0/+5 minutes. All exterior connectors shall have a mating connector, cover, or equivalent installed prior to exposure. After exposure, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI.

4.7.3.2.4 Washdown. The VMS modem shall be subjected to a washdown in accordance with 3.2.2.2.4 using a pressurized, hot water washer. Washer nozzle shall produce a fan or conical shaped spray pattern. Water shall be applied in a slow, sweeping motion over all surfaces for 2 minutes +1/-0 minute. Spray nozzle shall be held 6-8 inches from exterior surfaces of the unit. All exterior connectors shall have a mating cable, connector cover or equivalent installed prior to exposure. Following washdown, the unit shall be air dried for 2 hours ± 5 minutes at a temperature not to exceed 145°F. After exposure and drying, the VMS modem shall be tested per the quality conformance performance tests specified in Table VI.

4.7.3.3 First article sample inspection. The item shall be visually inspected (externally) for damage resulting in workmanship defects in accordance with 3.5.1. The covers shall be removed and the plug-in circuit card assemblies extracted. The circuit card assemblies and the item interior shall be visually inspected for damage resulting in workmanship defects in accordance with MIL-STD-454, Requirement 9.

4.7.4 Environmental stress screening (ESS). The manufacturing process shall be monitored for compliance to 3.3.

4.7.5 Workmanship.

4.7.5.1 General workmanship. General workmanship shall be in accordance with MIL-STD-454, Requirement 9. The LRU shall be visually inspected for legible identification markings specified in accordance with the applicable drawing.

4.7.5.2 Solder workmanship. Soldering workmanship shall be inspected in accordance with MIL-STD-2000.

4.7.6 Packaging inspection.

4.7.6.1 Submission. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product. Unless otherwise specified, sampling for examination shall be in accordance with 4.6.1.

4.7.6.2 Inspection to be performed.

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4.7.6.2.1 Level A. When Level A packaging is specified in the contract, samples, selected in accordance with Table VI, shall be inspected for the defects identified in Table VIII to verify compliance with section 5.

TABLE VIII. Level A packaging inspection provisions.

Requirement	Inspection Criteria	Method
Cleaning	Item not completely clean. Item damaged by cleaning process.	Visual Visual
Drying	Item not completely dry. Item damaged by drying process.	Visual Visual
Unit packing	Bubble wrap incomplete or not secured. Improper amount of desiccant. Desiccant contacting item. Heat seal test. Humidity indicator not properly placed. Unit container not as specified. Cushion material not as specified. Improper closure of unit container. Excessive looseness in unit container.	Visual Visual Visual 1/ Visual Visual Visual Visual Shake/Visual
Marking	Marking illegible, incorrect, or incomplete.	Visual

1/ Test in accordance with MIL-P-116, group B inspection.

4.7.6.2.2 Level C. When Level C packaging is specified in the contract, samples, selected in accordance with Table VI, shall be inspected for the defects identified in Table IX to verify compliance with section 5.

TABLE IX. Level C packaging inspection provisions.

Requirement	Inspection Criteria	Method
Unit packing	Unit container not as specified. Improper closure of unit container. Excessive looseness in unit container.	Visual Visual Shake/Visual
Marking	Marking illegible, incorrect, or incomplete.	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C as specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning. The assembly shall be cleaned in accordance with MIL-P-116, process C-1.

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5.1.1.2 Drying. The assembly shall be dried in accordance with MIL-P-116, as required.

5.1.1.3 Preservative application. Preservative shall not be applied.

5.1.1.4 Unit packs. Each assembly shall be individually unit packed in accordance with submethod IIb of MIL-P-116 in a bag in accordance with MIL-B-117, type 1, class F, style 1. To prevent puncturing of the bag, any protruding parts or sharp edges shall be cushioned with cushion material, flexible, cellular plastic film, class 2 per PPP-C-795 (bubble pack) prior to placement in the bag.

5.1.2 Level C. Unit pack shall be as required to safely transport and handle the VMS modem throughout the distribution cycle.

5.2 Packing. Unless otherwise specified by contract or purchase order, packing of the assembly shall be in accordance with Level A or C requirements of MIL-STD-2073.

5.3 Marking.

5.3.1 Standard marking. In addition to any special or other identification marking required by the contract, each unit and exterior container shall be marked in accordance with MIL-STD-129.

5.3.2 Special marking. In addition to the marking requirements of 5.3.1 and regardless of the level or type of packaging specified, special markings may also be required (see 6.2).

5.4 General.

5.4.1 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.7.6

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The VMS modem is intended for use as a component of the M109A6 Howitzer automatic fire control system.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Time frame required for submission of first article and number of units required (see 4.5).

d. Levels of preservation and packaging (see 5.1 and 5.2).

e. Permission for the contractor to optimize or tailor ESS and preparation of an ESS optimization or tailoring plan per MIL-STD-785 Task 301, if applicable (see 3.3).

f. SRU ESS requirements if applicable (see 3.3).

g. Special test equipment approval requirements (see 4.4).

h. Disposition instructions for nonconforming material (see 4.5.3 and 4.6.2).

i. Special marking requirements (see 5.3.2).

j. MIL-STD-2000 tailoring (see 3.6.2).

k. Pertinent first article information (see 6.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a preproduction sample, a first article sample, a first production item, and the number of items to be tested as specified in 4.5.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired and tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Submission of inspection equipment designs for approval. Contractor designs for final acceptance inspection equipment should be approved by the Government prior to fabrication or procuring the equipment. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, AMSTA-AR-QAC-F, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1423 in the contract.

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6.5 Subject term (key word) listing.

AFCS
Fire Control System, Automatic

6.6 EMI/EMP/Nuclear.

6.6.1 Electromagnetic radiation.

6.6.1.1 Electromagnetic interference/compatibility. The control of electromagnetic emission and susceptibility characteristics of the assembly shall be maintained in accordance with MIL-STD-461B. Electromagnetic compatibility requirements shall be maintained in accordance with MIL-E-6051D(1).

6.6.1.2 Electromagnetic radiation hazard effects. The assembly shall be maintained for safety and reliability in accordance with the Army electromagnetic radiation (EMR) criteria of the Adjutant General's Office (TAGO), listed in technical report MICOM-TR-RT-81-5.

6.6.1.3 Lightning effects. The assembly shall be maintained to be operationally unaffected after a near hit lightning strike (10 meters) when installed in an M109A6 Self Propelled Howitzer. Lightning and electrostatic hazards shall be as specified in MICOM-TR-RT-81-5.

6.6.1.4 Electromagnetic pulse effects/ Nuclear Survivability/ Vulnerability. The assembly shall be maintained to meet an EMP environment, when installed in an M109A6 Self Propelled Howitzer, as specified in the "Nuclear Survivability Criteria for Howitzer Improvement Program (HIP)," established by the U.S. Army Nuclear and Chemical Agency (USANCA) letter for MONA-NU to AMCPM-CAWS-W, Subject: Request for Nuclear Survivability Criteria, dated 15 Mar 85 (document CONFIDENTIAL). The assembly baseline survivability hardness levels are described in the reports:

a. Test Report, Preproduction Qualification Test (PQT) of Howitzer Improvement Program (U), Edwin K. Dunlap, Judith Ross, Nuclear Survivability Assessment Division, Nuclear Effects Directorate, White Sands, New Mexico, dated November 1989, TECOM Project No. 2-WE-200-HIP-029 (document SECRET).

b. Memorandum for Program Manager, Howitzer Improvement Program, ATTN: AMCPM-HIP, Dover, NJ, dated 14 August 1991. SUBJECT: Interim Report on Corrective Action Testing of M109A3E2 Howitzer (HI) Nuclear Survivability Assessment Program (NSA), TECOM Proj #2-WE-206-HIP-209 (U), Technical Risk Assessment, 6 Parts (U) (document SECRET).

6.7 Expanded definitions. The VMS modem accommodates vehicle speeds in the range of -80 to +80 miles per hour. Each output

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pulse of the VMS's odometer lines represents 0.165 feet of vehicle motion. Pulses generated by the VMS, 12927632, processed, and regenerated by the VMS modems are 800 microseconds in width. The pulse rate varies from 0 to 711 pulses per second (0 to 80 miles per hour). This information is used by the Dynamic Reference Unit (DRU), MIL-D-70789 (AR), in determining vehicle position.

The VMS modem incorporates loopback testing in which the frequency-encoded data generated by the VMS modem is fed back into its own frequency decoding circuits to allow isolation of faults to a single Line Replaceable Unit (LRU). The execution of these loopback tests by the Control Unit (CU) takes control of the BIT Command Lines away from the DRU, and causes normal Odometer Forward and Odometer Reverse data flow to be discontinued. Therefore, the loopback tests are performed when the DRU does not require odometer data (such as on power-up or after BIT has already detected an error in the VMS or VMS modems).

There are two identical VMS modem assemblies, 12561737, in the M109A6 AFCS. The function of the chassis-side VMS modem is to encode and modulate the odometer forward and odometer reverse outputs of the VMS unit into unique frequencies. These frequencies will be transmitted over the slip rings to the cab-side VMS modem. The cab-side VMS modem decodes the frequencies to their original state for use by the DRU. Also, the cab-side VMS modem will encode and modulate the BIT commands of the DRU into unique frequencies to be transmitted over the slip rings and decoded by the chassis-side VMS modem. The cab-side VMS modem is additionally controlled by the BIT command and test outputs of the Control Unit (CU).

The VMS modem which uses unconditioned +28V power will also provide regulated voltages needed for the modem itself and the VMS.

6.7.1 VMS modem interfaces. A block diagram of the VMS modems as connected in the AFCS is shown in Figure 8. A description of the functions and interfaces of the modem is given in the following paragraphs.

6.7.1.1 Chassis-side VMS modem. The chassis side VMS modem provides the functions as described below when connected as part of the AFCS as shown in Figure 8.

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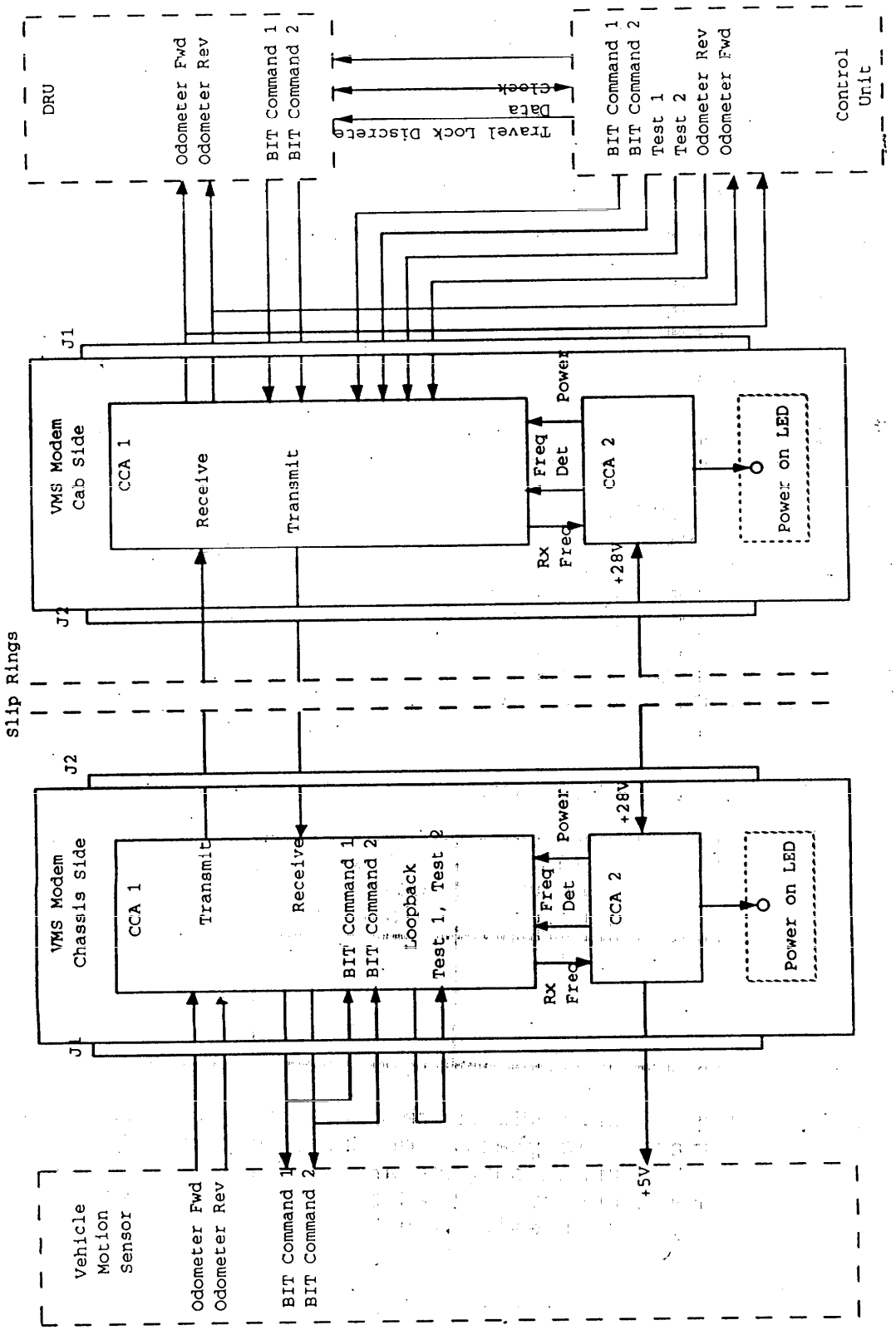


FIGURE 8. VMS modem block diagram.

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The chassis side VMS modem receives Odometer Forward and Odometer Reverse pulses from the VMS. Each pulse gates a signal of a unique frequency which is transmitted to the cab side VMS modem via a turret slip ring. When no pulses are received by the chassis side VMS modem, a 6.25 kHz signal is continuously transmitted. The presence (1) or absence (0) of a pulse vs. the frequency transmitted is shown in Table X.

TABLE X. Chassis-side VMS modem transmit frequencies.

Odometer Forward (input)	Odometer Reverse (input)	Square-Wave Frequency (output)
0	0	6.25 kHz
0	1	100 kHz
1	0	50 kHz
1	1	25 kHz

The chassis side VMS modem also receives a frequency encoded signal from the cab side VMS modem via a slip ring for the purpose of BIT testing and diagnosis of a fault to either the VMS or the chassis side VMS modem. Depending on the frequency of the signal received, the BIT Command 1 or 2 output states of the chassis side VMS modem are changed. A frequency of 6.25 kHz is continuously received when BIT is not commanded. The relationship of frequencies to BIT Command state, while the loopback output is 0, is shown in Table XI.

TABLE XI. Chassis-side VMS modem receive frequencies.

Square-Wave Frequency (input)	BIT Command 1 (output)	BIT Command 2 (output)	Loopback (output)
6.25 kHz	0	0	0
100 kHz	0	1	0
50 kHz	1	0	0
25 kHz	1	1	0

When a frequency of 12.5 kHz is received, the chassis side VMS modem is put in a special loopback mode for approximately 0.1 second. Since the BIT Command outputs are tied to BIT Command inputs and the Loopback output is tied to Test 1 and Test 2 inputs via external cabling, the chassis side VMS modem can simulate odometer pulses during this loopback time. The simulation is possible because setting Test 1 and Test 2 to 1's by the loopback output deselects the Odometer Forward and Reverse inputs internally within the modem and selects instead the BIT Command 1 and 2 inputs in order to simulate odometer pulses. The chassis

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side VMS modem input/output relationships for loopback mode are shown in Table XII.

TABLE XII. Chassis-side VMS modem loopback.

Square-Wave Frequency (input)	BIT Command 1 (output/input)	BIT Command 2 (output/input)	Loopback (output)	Square-Wave Frequency (output)
12.5 kHz	0	0	1 1/	6.25 kHz
6.25 kHz	0	0	1 2/	6.25 kHz
100 kHz	0	1	1 2/	100 kHz
50 kHz	1	0	1 2/	50 kHz
25 kHz	1	1	1 2/	25 kHz

1/ Loopback is in the 1 state for a duration of 0.1 sec.

2/ BIT Command 1 and 2 select the output frequency only while loopback is in the 1 state.

The chassis side VMS modem also supplies a regulated +5 Vdc to the VMS. This power source is not damaged and continues to operate after exposure of the input voltage to the following:

a. Voltage surges and spikes as defined in MIL-STD-1275A(AT).
b. Low voltage.
c. Overvoltage.
d. Reverse polarity.

6.7.1.2 Cab side VMS modem. The cab-side VMS modem provides the functions as described below when connected as part of the AFCS as shown in Figure 8.

The cab side VMS modem receives a frequency encoded signal from the chassis side VMS modem via a turret slip ring. Depending on the signal frequency detected by the cab side VMS modem, an 800 microsecond pulse is generated on the Odometer Forward or Odometer Reverse outputs as listed in Table XIII.

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TABLE XIII. Cab side VMS modem receive frequencies.

Square-Wave Frequency (input)	Odometer Forward (output)	Odometer Reverse (output)
6.25 kHz	0	0
100 kHz	0	pulse
50 kHz	pulse	0
25 kHz	pulse	pulse

The cab side VMS modem also receives BIT Commands 1 and 2 from two different sources in the AFCS (the DRU and the CU). The source which controls BIT testing is selected by the AFCS's CU via the state of the Test 1 and 2 inputs to the modem.

When the Test 1 and Test 2 inputs are both set to 0, the BIT Commands inputs which normally come from the DRU in the AFCS cause a signal to be transmitted with the appropriate frequency. The frequency of the transmitted signal versus the state of the BIT Command inputs is shown in Table XIV.

TABLE XIV. Cab side VMS modem transmit frequencies.

BIT Command 1 DRU (input)	BIT Command 2 DRU (input)	Square-Wave Frequency (output)
0	0	6.25 kHz
0	1	100 kHz
1	0	50 kHz
1	1	25 kHz

When the Test 1 input is set to 0 and the Test 2 input is set to 1, the BIT Command inputs normally from the AFCS's CU gate a signal with the appropriate frequency to the receive function of the cab side VMS modem via an internal relay. The detected frequency is then transmitted by the modem as 800 microsecond odometer pulse outputs. This is summarized in Table XV.

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TABLE XV. Cab side VMS modem BIT.

BIT Command 1 CU (input)	BIT Command 2 CU (input)	Odometer Forward (output)	Odometer Reverse (output)
0	0	0	0
0	1	0	pulse
1	0	pulse	0
1	1	pulse	pulse

When the Test 1 input is set to 1 and the Test 2 input is set to 0, the cab side VMS modem transmits a 12.5 kHz signal to the chassis side VMS modem via a slip ring for the duration of the time that the Test inputs remain in this state.

When Test 1 and 2 are both set to 1, the CU BIT Command inputs select which frequency is transmitted from the cab side VMS modem to the chassis side VMS modem via a slip ring. The frequencies transmitted vs. BIT Command states are shown in Table XVI.

TABLE XVI. Cab side VMS modem transmit frequencies.

BIT Command 1 CU (input)	BIT Command 2 CU (input)	Square-Wave Frequency (output)
0	0	6.25 kHz
0	1	100 kHz
1	0	50 kHz
1	1	25 kHz

6.7.2 Physical description. The VMS modem consists of a single, sealed, enclosed housing and the following Circuit Card Assemblies (CCAs):

a. VMS modem 2 CCA, 12561839
b. VMS modem 1 CCA, 12561843

6.7.2.1 VMS modem 1 CCA: As shown in Figure 8, the VMS modem 1 CCA interfaces with the DRU and Control Unit (CU), or with the VMS (depending on whether it is a cab side or chassis side modem, respectively) using differential signals. It also interfaces with the vehicle slip rings, receiving and transmitting frequency encoded signals. Internally to the VMS modem, the VMS modem 1 CCA interfaces with the VMS modem 2 CCA, providing the received/loopback signal (the R_x frequency in Figure 8) and accepting the frequency detect signals and power.

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The VMS modem 1 CCA contains the logic which performs the following functions:

a. Reception of all differential inputs to the VMS modem.

b. Generation and selection of the frequencies to be transmitted by the VMS modem in normal operation.

c. Wraparound of these frequencies in BIT mode via a loopback relay.

d. Conditioning of the frequency detect inputs from the VMS modem 2 CCA.

e. Generation of all differential outputs of the VMS modem.

6.7.2.2 VMS modem 2 CCA. As shown in Figure 8, the VMS modem 2 CCA interfaces with the vehicle slip rings, receiving vehicle unconditioned power. The VMS modem 2 CCA also interfaces to the VMS supplying regulated +5 Vdc. Internal to the VMS modem, the VMS modem 2 CCA interfaces with the VMS modem 1 CCA, accepting the received/loopback signal (R_x frequency in Figure 8) and providing the frequency detect signals. It also sources power for the VMS modem 1 CCA, including the opto-isolator interface, relay (internal to CCA 1), and the power on LED.

The VMS modem 2 CCA contains the circuitry which performs the following functions:

a. Opto-isolation of the received input frequency.

b. Detection of specific frequencies including 12.5 kHz, 25 kHz, 50 kHz, and 100 kHz.

c. Conditioning of the vehicle input power and generation/regulation of output power.

6.7.3 Connector description.

6.7.3.1 Slip Ring connector. The VMS modem interfaces with the Slip Rings via the J2 connector. The frequency-encoded transmit and receive data, as well as the unconditioned +28 Vdc and ground, are brought to the VMS modem through this connector. The cab-side J2 connector and the chassis-side J2 connector pin assignments are given below:

Cab Side J2		Chassis Side J2	
A	+28 Vdc uncond. power(in)	A	+28 Vdc uncond. power (in)

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B	+28 Vdc uncond. return (in)	B	+28 Vdc uncond. return (in)
C	Cab-to-chass BIT data (out)	C	Chass-to-cab odo. data (out)
D	Chass-to-cab odo. data (in)	D	Cab-to-chass BIT data (in)
E	Shield continuity test	E	Shield continuity test

6.7.3.2 DRU/VMS connector. The VMS modem interfaces to the DRU and the CU, or to the VMS, via the J1 connector. The BIT COMMAND lines, the ODOMETER FORWARD and REVERSE lines, and the LOOPBACK and TEST lines are brought into and out of the VMS modem via this connector. A voltage of +5 Vdc is supplied to the VMS by the chassis side VMS modem through this connector. The cab-side J1 connector and the chassis-side J1 connector pin usages as selected by external cable are shown below:

	Cab Side J1		Chassis Side J1
A	BIT comm 2 hi (DRU) (in)	A	Odo. reverse hi (VMS) (in)
B	BIT comm 2 lo (DRU) (in)	B	Odo. reverse lo (VMS) (in)
C	BIT comm 1 hi (DRU) (in)	C	Odo. forward hi (VMS) (in)
D	BIT comm 1 lo (DRU) (in)	D	Odo. forward lo (VMS) (in)
E	Reserved	E	BIT comm 1 hi (VMS) (out)
F	Reserved	F	BIT comm 1 lo (VMS) (out)
G	Reserved	G	BIT comm 2 hi (VMS) (out)
H	Reserved	H	BIT comm 2 lo (VMS) (out)
J	Reserved	J	+5V power (out)
K	Reserved	K	+5V power (out)
L	BIT comm 1 hi (CU) (in)	L	BIT comm 1 hi (VMS) (in)
M	BIT comm 1 lo (CU) (in)	M	BIT comm 1 lo (VMS) (in)
N	BIT comm 2 hi (CU) (in)	N	BIT comm 2 hi (VMS) (in)
P	BIT comm 2 lo (CU) (in)	P	BIT comm 2 lo (VMS) (in)
R	Reserved	R	Digital return
S	Reserved	S	Digital return
T	Reserved	T	Loopback hi (out)
U	Reserved	U	Loopback lo (out)
V	Odometer forward hi (out)	V	Reserved
W	Odometer forward lo (out)	W	Reserved
X	Odometer reverse hi (out)	X	Reserved
Y	Odometer reverse lo (out)	Y	Reserved

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Z	Test 1 hi (CU) (in)	Z	Loopback hi (in)
a	Test 1 lo (CU) (in)	a	Loopback lo (in)
b	Test 2 hi (CU) (in)	b	Loopback hi (in)
c	Test 2 lo (CU) (in)	c	Loopback lo (in)

6.7.3.3 Shield grounding. Inner and outer connector shields are grounded to the chassis through the backshell of the connector. The total impedance of the shields to chassis is less than 2.5 milliohms from DC to 10 MHz.

6.7.4 Internal box wiring. The VMS modem drawing, 12562462, shows the internal box connections between the J1 and J2 connectors, the VMS modem CCA1 and CCA2 SRUs, and the Power-On LED.

6.8 Changes from previous issue. The previous issues were for ARDEC internal use only. These issues were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is the initial public release for unlimited distribution.

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